Department of CHEMISTRY
Clemson University

CH 1010 General Chemistry
Course Syllabus for Spring 2025

Information about CH 1010

Catalog Course Description
Introduction to the elementary concepts of chemistry through classroom and laboratory experience. Emphasizes chemical reactions and the use of symbolic representation, the mole concept and its applications and molecular structure. Credit toward a degree will be given for only one of CH 1010 and CH 1050. This course has a corequisite laboratory course CH 1011.

Course Overview
Chemistry encompasses the vast realm of the physical world. It is intimately involved in all the physical and life sciences and is a foundation of many engineering fields. In CH 1010, you will learn how the structure of atoms and molecules determines the behavior of the world and universe in which we live (structure-property relationships).

You will be required to learn the language of chemistry and understand basic chemical concepts. Simple memorization will not be sufficient for success in the course; instead, you will be required to demonstrate the ability to think critically in applying your understanding of the course material.

Learning chemistry cannot be done by simply attending class. Working on your own outside of your formal lecture time will be required. Practice with the lecture content is an important part of the learning process.

The purpose of this syllabus is to outline the policies and procedures for CH 1010. As there are multiple sections/instructors for the course, additional section specific details can be found in the specific section syllabus published by each individual instructor. Information relevant to this course and to your section can be found on Canvas.

Co-requisite (Lab)
CH 1010 has a co-requisite course CH 1011 (General Chemistry Laboratory) which will *begin the week of January 27*. Look for an email from the lab coordinator, Dr. Perla Saucedo, psauced@clemson.edu, for key information regarding CH 1011.

- A separate syllabus will be issued for CH 1011.
- If you are repeating CH 1010 and previously earned a grade of **75 or above** in CH 1011, you may request that your previous CH 1011 score be used in the computation of this term's CH 1010 grade. The request should be made to Ms. Christy Lester, cmleste@clemson.edu. Be sure include your name, CUID number and the semester in which you took CH 1011. Should you be eligible, you will be moved into the exempt lab section and not required to attend lab. Note however, you will still be charged a lab fee for CH 1011, even though registered for the exempt section.

A student may NOT withdraw from CH 1010 without also withdrawing from CH 1011.

CH 1010 plus its companion lab, CH 1011, is a course which satisfies the General Education requirement for a natural science with a lab. The General Education competencies that are addressed in this course are:

- Natural Sciences: Students will demonstrate the process of scientific reasoning through experimental activity and critical comparison of their results to those predicted by accepted natural science principles.

Required Math Skills
This course does not have a prerequisite math requirement; however, a working knowledge of algebra and adequate problem-solving skills are assumed. You should be able to do the following (some of these will be reviewed briefly, but if you cannot do these, you should work especially hard now, or consider delaying Chemistry until you build your background skills). This background knowledge is expected on all exams!

- Use mathematical terms and equations including algebra, exponential numbers, logarithms, ratios and proportions.
- Use significant figures.
- Interpret and work with numbers in scientific notation.
- Convert between English - metric (SI) units.
- Make and read graphs.
- Interpret word problems.

Class Modality
Currently Sections 1-6 of CH 1010 are in-person with physical attendance of students expected. However, the course modality may be adjusted for short periods of time as required by the university or your instructor. Such adjustments will be announced with as much lead-time as possible and through Canvas, email and/or in-class announcements.

In the case that your instructor is unable to physically be present for one or more lectures, your instructor will communicate to you how the course will proceed. Expect the class to move temporarily to a virtual (online) format; look for communication from your professor by email.

Section 400 is online asynchronous. However, exams will not be virtual; exams must be taken in-person, on the schedule dates in a proctored environment. Although this section does not meet regularly except for exams, students are expected to interact with the course content on a daily basis. The instructor will be available for consultation.

Required Materials
- *Mastering Chemistry* with the accompanying etextbook *Chemistry, Structure and Properties* 3rd edition by Nivaldo J. Tro. There are two options for purchasing access.
    - Purchase directly from Pearson via the "Access Pearson" Canvas course navigation link. You will need a credit card.
    - Purchase an access code from the Clemson bookstore.
      Note that these are the only options. An access code purchased through a third party such as Amazon will not work for this course.

  There is a 14-day trial access which can be used to create an account and begin homework assignments until access is purchased.

  Pearson offers technical support during the semester for *Mastering Chemistry*: https://support.pearson.com/getsupport/s/ Alternatively, there is a 24/7 live chat option available by going to the bottom of the Pearson landing page and clicking on "Contact Us". Always make sure to note the incident number given by support.
- A computer capable of running all aspects of the Canvas learning management system, Zoom, Respondus Lockdown browser and Respondus Monitor throughout the entirety of the semester. Lack of having a computer or computer issues of any kind are not valid excuses for failing to complete an assignment. If not connected to the Clemson Wi-Fi network, a student is expected to have access to a high-speed internet connection. To ensure access to the campus resources when not on campus, you may want to use the CU Virtual Private Network (CUVPN). Information on downloading, installing and connecting can be found on the CCIT webpage.
    - If you are having hardware or software problems, CCIT's Service Desk may be able to help you. Contact them by

    - Email: ITHELP@clemson.edu
    - Call or text: (864) 656-3494
    - Live chat at ccit.clemson.edu.
    - Visit the help desk is located in Cooper Library.
- Non-programmable, **non-graphical** scientific calculator.
    - Possible models include but are not limited to:
        - *Casio*: all fx-115 models;
        - *HP*: 33 or 35 models;
        - *TI*: 30X or 36X models.
    - If you have a calculator that you think may be acceptable, please ask for clarification. A cell phone is not an acceptable calculator.

Other Materials

- Some instructors use a class response system which requires a separate purchase. Your instructor will provide additional information if access to the system is required.
- An optional math review can be obtained from the bookstore: *Math Survival Guide: Tips for Science Students*, J. Appling and J. C. Richardson (2nd Ed. Wiley, 2009). Note that appendix 1 of the textbook has a math review. Cost-wise your textbook is the better option.

Lecture Content Schedule for CH 1010 Spring 2025

Note that this schedule is tentative and subject to change.

Lecture Notes Chapter	Tro Chapter	TOPIC
Chap. 1	E	Properties and Measurement
Chap. 2	1	Atoms and Elements
Chap. --	--	Intro to Energy
Chap. 3	2	Waves, Light, and Electrons
Chap. 4	3	Electron Configurations on Atoms
EXAM 1		**Monday February 10th (7:30 pm)**
Chap. 5	3	Periodic Trends
Chap. 6	4-5	Molecules and Salts
Chap. 7	5-6	Molecular Structure
EXAM 2		**Monday March 10th (7:30 pm)**
Chap. ---	11	Intermolecular Forces
Chap. 8	7	Stoichiometry
Chap. 9	8	Aqueous Solutions
Chap. 10	9	Energy
EXAM 3		**Monday April 21st (7:30 pm)**
Chap. 11	10	Gases
FINAL EXAM		**Wednesday April 30th (11:30 am)**

	Sections of Tro	Topic
List of the Main Exam 1 Topics	E.1 – E.9 1.1-1.10 2.1-2.6 3.1-3.7	• Introduction to CH 1010 • Units including density. • Significant figures in measurements & calculations • Conversion factors • The Atomic Theory of matter • Classifications of matter • The particulate nature of matter • Phases of matter • Introduction to energy • The structure of the atom • Isotopes • Atomic mass • The mole • Wave properties of electromagnetic radiation • The photoelectric effect • Atomic spectra • The quantum mechanical model of the atom • Quantum numbers • Shapes of orbitals • The periodic table • Electron configurations • Anomalous electron configurations • Electron configurations of ions • Effective nuclear charge • Atomic size
List of the Main Exam 2 Topics	3.7 – 3.9 4.1 – 4.11 5.2 – 5.10 9.11 6.1 – 6.5	• Other periodic properties • Compounds, molecular versus ionic • Polyatomic ions • The compound formula • Naming simple compounds • Determining empirical & molecular formulas • Lattice energy • Bond energies • Electronegativity and polar bonds • Representing molecules with Lewis Structures • Resonance • Formal Charges • Electron geometry • Shapes of molecules and polyatomic ions • Valence Bond Theory • Hybrid orbitals
List of the Main Exam 3 Topics	11.1 – 11.3 7.1 – 7.5 9.1-9.10 8.1-8.7, 8.9	• Molecular polarity • Intermolecular forces • Structure property relationships • Balancing chemical reactions • Stoichiometric calculations • Yields of reactions • Systems and state functions • Internal energy of a system • Work • Specific heat • Enthalpy • Calculating an enthalpy change using Hess's Law • Calculating an enthalpy change using bond dissociation energies

		• Calculating an enthalpy change using enthalpies of formation • Molarity • Calculations involving molarity • Strong, weak and non-electrolytes • Solubility rules • Writing molecular, complete ionic and net ionic reactions • Precipitation reactions • Acid-base neutralization reactions • Oxidation-reduction reactions •
List of the Last Week Topics	10.1 – 10.11	• Stoichiometric calculations with gases • Dalton's Law of partial pressures • Diffusion of gases • Real gas behavior

Other Important Dates:

- *First day of class*: January 8 for MWF sections and January 9 for TR sections.
- *MLK Holiday, no classes meet*: Jan. 20,
- *Last day to* withdraw from CH 1010 without the course appearing on your transcript: Wednesday, January 22.
- *Last day to drop* the course without a final grade: Wednesday, March 14. A grade of "W" will appear on your transcript.
- *Spring Break*: Mar. 17 (Mon.) – Mar. 21 (Fr.)
- Last class meeting: April 24 (TR) or April 25 (MWF).
- Grades will be issued no later than Wednesday May 7.

A student may NOT withdraw from CH 1010 without also withdrawing from CH 1011.

If you have questions or issues with regard to the course administration, see your instructor. For general questions, see Ms. Christy Lester in Hunter 219; telephone: (864) 656-3089, email: cmleste@clemson.edu. If necessary, see the Director of General Chemistry, Dr. Taylor in Hunter 265; email: dftay@clemson.edu.

Course grades will be apportioned as shown.

• Exams (three total)	45%
• Final Exam	20%
• Homework	5%
• Class Activities	9%
• Laboratory (CH 1011)	20%
• Skills for Chemistry Assessment	1%
• **Total**	**100%**

Final grades will be assigned based on the ranges shown. Do not assume that grades will be rounded or curved.

- 90 through 100 A
- 80 through 89 B
- 70 through 79 C
- 60 through 69 D
- below 60 F

At the end of the semester as final grades are being assigned, DO NOT email your instructor asking that your average be rounded up to the next letter grade. Such emails will not be acknowledged. Assume that a "round-up" of course averages will not take place, i.e. expect a course average of 79.5 to result in a CH 1010 grade of "C."

Class Activities

What constitutes "Class Activities" will vary by instructor. See your instructor's *Section Syllabus* for more information.

Help with CH 1010

You may obtain extra help from the following resources.

- Your PAL Leader: See the later paragraph on peer-assisted learning via the Academic Success Center (ASC) on page 12.
- Tutoring. See the later paragraph on this service provided by the ASC also on page 11.
- Your lecture Instructor: See your instructor's *Section Syllabus* for student hours, a.k.a. office hours.
- Form a study group with other students who are taking CH 1010. (They do not have to be in your section.) Working in small groups with others can really help.
- Help sessions may be offered periodically by instructors.

If you are having hardware or software problems, CCIT's Service Desk may be able to help you. Contact them by emailing ITHELP@clemson.edu, calling or texting (864) 656-3494, or starting a live chat at ccit.clemson.edu. The help desk is located in Cooper Library.

Exam Preparation

Exams are written to test students' conceptual knowledge and problem-solving ability rather than the ability to memorize material. Even if a student could memorize every single thing presented in the course, the student would be unlikely to perform well on exams unless he/she was also able to apply the concepts presented in the course. Practice is the best way to learn the material. Practice questions will be provided to help in preparation for exams. These practice questions will provide an idea of the types of questions that might appear on the actual exam. Do not expect exam questions to duplicate practice questions.

All exams should be considered comprehensive; previously covered material will be tested. Each exam will cover the material listed in the objectives posted on Canvas, or as modified in class.

Attempts will be made to cover most of these in class. Covered or not, these objectives will be subject to examination.

Exam Administration
Exams 1, 2 & 3 are given at 7:30 P.M. on the following Monday nights:
Exam 1: February 10 Exam 2: March 10 Exam 3: April 21

Exam administration details, including location, will be communicated to you by your instructor about one week prior to the exam. You are required to be physically present on-campus in your assigned location for each exam. You will be expected to bring your own laptop capable of running LockDown Browser to this exam. Students requiring testing accommodations must schedule time with the University Testing & Educational Center (UTEC) at least one week prior to the exams date and three weeks prior to the final exam.

Exams will be administered using Canvas. Exam dates cited earlier in this document. LockDown Browser and possibly Respondus Monitor will be required during each exam. Questions must be answered in the order presented; once an answer is submitted returning to a previous question will not be allowed. When you have completed your exam, you will show the Canvas submission page to an exam proctor and confirm your identity with an appropriate ID. You may be required to initial a sign-out sheet to confirm your presence at the exam. Failure to initial the sign-out sheet may lead to a student being marked as "absent" and/or an academic integrity violation charge. Additional details regarding exam administration will be provided closer to the first exam.

A student is required to be physically present in a proctored environment for each exam. This requirement also applies to students in the online section. Requests for a virtual exam will be denied.

The calculator that you use for your exams must be a scientific calculator. A graphing/programmable calculator may NOT be used, nor may a cell phone or other mobile device. See the Required Materials section on page 3.

Any type of communication or electronic device (e.g. cell phone, Apple watch, Fitbit, iPod, headphones, etc.) is prohibited during exams; such devices should be stored out of sight or left at your residence. Any student observed with a communication/electronic device will be subjected to a charge of academic dishonesty.

Each exam will cover the material listed in the objectives posted on Canvas or as modified in class. Attempts will be made to cover these in class, however, covered or not, these objectives will be subject to examination.

Final Exam
The final exam is a required part of a student's CH 1010 grade. There will be no exemptions from the final exam; all students must take the final exam in order to earn a passing grade. The Final Exam in CH 1010 will be cumulative over all topics covered during the semester. Note that the Final Exam in CH 1020 will be cumulative over two semesters, covering material from both CH 1010 and CH 1020

Exam Replacement Policy:
The Final Exam grade may substitute for your lowest exam score of the semester if it is to your advantage. Scores for unexcused exam absences will not be replaced. If your Final Exam score is lower than all scores on exams 1-3, then no substitution will take place.

Final Exam scheduling issues must be resolved at least two weeks prior to finals week. A student who has more than two exams in one calendar day may request that one of the exams be taken at a different time and date. The registrar has established a priority as to which course must provide an alternative exam time. The CH 1010 Final Exam has top priority and cannot be

rescheduled. See the registrar's webpage for additional information, https://www.clemson.edu/registrar/student-menu/examinations.html, then look under "Students", "Examinations" and Spring Exam Schedule."

Missed Exam Policy

Exams 1-3

Should a student be unable to take a scheduled exam due to an emergency situation, such as isolation/quarantine, illness or a death in the family, the student should contact their instructor as soon as possible, providing documentation of the emergency. It is recommended that the Notification of Absence tool, found in Canvas, be used with a follow-up email to communicate with the instructor. If the absence has been classified as "excused" the student will be contacted to schedule a time to take the make-up exam. Personal travel (such as weddings, vacations, etc.) does not constitute an emergency and does not qualify for a make-up exam. Be aware that the make-up exam will have different questions which will not be easier than the regular exam.

Make-up exams will be administered in-person; there is no option for taking the make-up exam virtually. The make-up exams for Spring 2025 will be held the Friday after the scheduled exam beginning at 4:00 p.m. in Hunter 100. The dates are February 14th, March 14th and April 25th.

Final Exam

Should you be unable to take the final exam due to an emergency, contact your instructor prior to the exam time, providing documentation. A make-up will be scheduled, and the student will receive an Incomplete (I) grade in the course. The expectation is that the make-up exam will be completed within 30 days of the beginning of the spring semester. If you are taking CH 1020, the make-up final exam must be completed prior to the first day of classes of CH 1020. If you fail to make up the final exam, your grade of 'I' will automatically revert to a grade of 'F'.

Grade Appeals/Conflict Resolution

Should you believe that an error was made in the grading of your exam, you must register a written appeal with your instructor within 10 days of the exam, including the missed question(s), and your explanation of what you believe the error to be.

Testing Accommodations for Exams

Students who are entitled to testing accommodations through Student Accommodation Services (SAS) will need to schedule a time with the University Testing & Education Center (UTEC) to take all CH 1010 exams. The exam must be scheduled for the date noted previously – no late exams allowed; exams should start at or after 8:00 a.m. and must be finished by 6:30 p.m. the same day.

Scheduling must take place at least a week prior to the exam date, preferably earlier. Contact your instructor for additional details. A Final Exam should be scheduled by, at the latest, April 4. Be aware that testing space at the UTEC is at a premium during final exams week; priority is given to students who register early.

Schedule all three exams and the final as soon as your accommodation is approved!

Course Policies

Attendance Policy

The academic resources of Clemson University are provided for the intellectual growth and development of students. Class attendance is critical to the education process; therefore, students should attend scheduled courses regularly if they are to attain their academic goals. The university attendance policy can be found in the "Academic Regulations" section of the *2023-2024 Undergraduate Catalogue*.

Each CH 1010 instructor will have an attendance policy for their section. It is the responsibility of the student to make direct and timely contact with the course instructor regarding an absence, scheduled or otherwise. Documentation is expected. The required method of communication regarding an absence is the Notification of Absence tool found in Canvas. Additional communication via email is also encouraged.

The **Notification of Absence (NOA)** tool in Canvas allows students to quickly notify instructors (via an email) of an absence from class and provides for the following categories: court attendance, death of immediate family member, illness, illness of family member, injury, military duty, religious observance, scheduled surgery, university function, unscheduled hospitalization, other anticipated absence, or other unanticipated absence. The notification form requires a brief explanation, dates and times. Based on the dates and times indicated, instructors are automatically selected, but students may decide which instructors will receive the notification. This does not serve as an "excuse" from class. It is a request for an excused absence and students are encouraged to discuss the absence with instructors, as the instructor is the only person who can excuse an absence. If students are unable to report the absence by computer, they may reach the Office of Advocacy and Success. Students with excessive absences who need academic or medical assistance can also contact the Office of Advocacy and Success.

The Notification of Absence tool can be found in Canvas by going to the menu on the left and clicking on "help" which is the question mark in a circle icon.

A student with an excessive number of absences may be withdrawn (dropped) from CH 1010 at the discretion of the course instructor. An excessive number of absences is considered to be more than a week of missed classes, not necessarily consecutive.

Clemson University Policies

Policy on Canceled Lectures
If your lecturer is more than 15 minutes late, you may assume that the lecture will be canceled for the day. However, you are still responsible for the material that would have been reviewed during this period. It would be to the student's advantage to utilize this period reading the missed material from the current chapter.

Any assignments due at the time of a class cancellation will be due the next class meeting unless contacted by your instructor. Any extension or postponement of assignments or exams must be granted by the instructor via email or Canvas within 24 hours of the cancellation.

Any exam that is scheduled at the time of a class cancellation due to inclement weather, power outage, etc. will be given at the earliest possible date, probably the next week. Details will be communicated by your instructor through Canvas and/or your Clemson email.

Academic Continuity
Clemson has developed an Academic Continuity Plan for academic operations. Should university administration officially determine that the physical classroom facility is not available to conduct classes, class will be conducted in a virtual (online) form. The university issues official disruption notifications through email, website, text notification and Social Media. When notified, use one of the following links to navigate to Clemson Canvas where you will find important information about how we will conduct class:

- Primary access link: http://www.clemson.edu/canvas
- Secondary access link, if needed: https://clemson.instructure.com/
- You can also use the Canvas Student App. Visit the downloads page for this app.

Course activities will occur through the Canvas course.

Any exam that was scheduled at the time of a class cancellation due to inclement weather will be given at the next class meeting unless contacted by the instructor. Any assignments due at the time of a class cancellation due to inclement weather will be due at the next class meeting unless contacted by the instructor. Details will be communicated by your instructor through Canvas and/or your Clemson email as soon as possible, likely within 24 hours.

Emergency Preparedness Statement
Emergency procedures have been posted in all buildings and on all elevators. Students should review these procedures for their own safety. All students and employees should be familiar with guidelines from Clemson University Public Safety.

Clemson University is committed to providing a safe campus environment for students, faculty, staff, and visitors. As members of the community, we encourage you to take the following actions to be better prepared in case of an emergency:

1. Familiarize yourself with all possible exits, safer locations, and other key information on the emergency evacuation maps in this building, and those that you visit regularly.
2. Make a plan for how you would Run, Hide, and Fight in case of an active threat in this building, and those that you visit regularly. For example:
   - Run – what are all the possible exits in this building, and the routes to them?
   - Hide – what are the potential hiding locations in this room and building that are out of sight of doors and windows, how do you lock the door(s), how would you barricade the door(s) and windows, where do you turn off the lights?
   - Fight – What tools are available in this room and building, should you have to fight?
3. Ensure you are signed up for emergency alerts. Alerts are only sent when there is a potential threat to safety, a major disruption to campus services, and once-monthly tests.
4. Download the Rave Guardian app to your phone (https://www.clemson.edu/cusafety/cupd/rave-guardian/)
5. Learn what you can do to prepare yourself for the hazards that affect our locations (http://www.clemson.edu/cusafety/EmergencyManagement/)

Academic Integrity Policy
"As members of the Clemson University community, we have inherited Thomas Green Clemson's vision of this institution as a "high seminary of learning." Fundamental to this vision is a mutual commitment to truthfulness, honor, and responsibility, without which we cannot earn the trust and respect of others. Furthermore, we recognize that academic dishonesty detracts from the value of a Clemson degree. Therefore, we shall not tolerate lying, cheating, or stealing in any form."

When, in the opinion of a course instructor, there is evidence that a student has committed an act of academic dishonesty, that instructor will make a formal written charge of academic dishonesty to the Associate Dean of Undergraduate Studies. The instructor is not required to discuss the charge with the student. See the Undergraduate Academic Integrity Policy website for additional information and the current catalogue for the policy.

Examples of academic integrity violations include but are not limited to (i) reporting another student's answers as your own, (ii) being observed with a cell phone during the exam, (iii) accessing external websites during an exam or quiz, (iv) sharing the password of an exam or quiz (v) posting course material on an external website such as "Course Hero" or "Chegg" (vi) accessing or possessing any material not expressly permitted during an exam or quiz, such as crib sheets, notes, books, (vii) taking a picture of or otherwise copying an exam question without permission to do so, (viii) collaborating with unauthorized persons on exams, quizzes and/or homework assignments, (ix) giving unauthorized assistance to another student taking an exam or quiz, (x) submitting materials and responses generated using artificial intelligence without attribution (xi) taking an exam in a location other than that assigned for a student.

Accessibility Policy
Clemson University values the diversity of our student body as a strength and a critical component of our dynamic community. Students with disabilities or temporary injuries/conditions may require accommodations due to barriers in the structure of facilities, course design, technology used for curricular purposes, or other campus resources. Students who experience a barrier to full access to this class should let the instructor know and are encouraged to request accommodations through SAS (Student Accessibility Services) as soon as possible. To request accommodations through SAS, please see this link: https://www.clemson.edu/academics/studentaccess/register.html.

Students who are entitled to receive an accommodation through SAS are expected to meet individually and/or virtually with their chemistry instructor to discuss necessary accommodations. This meeting should be scheduled as early as possible in the semester so that accommodations can be made in a timely manner. It is the student's responsibility to schedule the meeting and ensure that accommodations are adequate. A notice of accommodation that is received less than a week prior to a scheduled exam may not qualify a student for accommodations due to lack of adequate lead time required to institute the expected accommodations.

Title IX Statement Regarding Non-Discrimination
Clemson University is committed to a policy of equal opportunity for all persons and does not discriminate on the basis of race, color, religion, sex, sexual orientation, gender, pregnancy, national origin, age, disability, veteran's status, genetic information or protected activity in employment, educational programs and activities, admissions and financial aid. This includes a prohibition against sexual harassment and sexual violence as mandated by Title IX of the Education Amendments of 1972. This Title IX policy is located on the Access and Equity website. Ms. Alesia Smith is the Clemson University Title IX Coordinator, and the Assistant Vice President of Equity Compliance. Her office is located at 223 Brackett Hall, 864-656-3181 and her email address is alesias@clemson.edu. Remember, email is not a fully secured method of communication and should not be used to discuss Title IX issues.

Copyright
Materials in courses are copyrighted. They are intended for use only by students registered and enrolled in a particular course and only for instructional activities associated with and for the duration of the course. They may not be retained in another medium or disseminated further. They are provided in compliance with the provisions of the Teach Act. Students must seek permission from instructors to record any class activity, including lectures, discussions, and presentations. Students should be reminded to refer to the Use of Copyrighted Materials and "Fair Use Guidelines" policy on the Clemson University website. Additional information can be found here.

Posting or sharing course related content with a third party such as Chegg, Course Hero, ClutchPrep, etc. is a violation of copyright.

Student Support

The Academic Success Center (ASC) offers a variety of free learning and success services for all undergraduate students that are designed to equip students with strategies and resources they can use to become a more confident, independent and skillful learner.

- Peer Tutoring – students can expect a 1:1 meeting with a trained undergraduate peer leader (who made an A or B in the course and was recommended by a faculty member) during which the student can share specific questions they have about course content with the tutor focused on helping the student, through questioning techniques and identification of helpful learning strategies, master course concepts. Tutors do not help with homework or other class assignments. (Linked to a course)

- Peer-Assisted Learning (PAL) – students can expect collaborative and active group learning and study sessions focused on mastery of course content and learning strategies that are facilitated by a trained undergraduate peer leader (who made an A or B in the course and was recommended by a faculty member). PAL leaders do not help with homework or other class assignments. (Linked to a course)
- Academic Coaching – students can expect a 1:1 meeting with a trained professional academic coach during which the coach helps students see themselves, their skills, and their study habits from a fresh perspective through one-on-one sessions focused on learning and personal success strategies.
- Success Strategy Workshops – students can expect 30 - 45-minute workshops on college success skills, time management and organizational skills, test-taking strategies, study strategies, finals preparation, life skills, and academic resources.

The Class of 1956 Academic Success Center building is located in the center of campus adjacent to Cooper Library and the Watt Family Innovation Center.

Student Learning Outcomes for CH 1010

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1. Demonstrate an understanding of the organization of the periodic table. Identify elements as metals, non-metals, metalloids and Noble gases.
2. Use the mole concept to interconvert among mass, moles, number of molecules, number of atoms, volume of gas (using gas laws), density and molarity.
3. Explain what is meant by the wave-particle duality of both light and matter.
4. Predict relative magnitudes of the elemental properties of electronegativity, electron affinity, ionization energy, and ionic or covalent radius from the position in the periodic table.
5. Use the periodic table to predict the ionic charge of an element in a compound and write formulas of chemical compounds using ionic charges.
6. Classify simple compounds as being molecular or ionic based on the compound's composition.
7. Explain the continuum that exists for bonding in compounds;
   ionic → polar covalent → non-polar covalent.
8. Draw Lewis structures for covalent compounds.
9. Use VSEPR concepts and Lewis structure to predict: a) bond angles, b) electron geometry, c) molecular shape and d) hybridization.
10. Use Valence Bond Theory and Molecular Orbital Theory to describe the bonding between atoms in a molecule.
11. Distinguish between intramolecular and intermolecular forces of attraction.
12. Identify the types of intermolecular forces and use them to predict properties of liquids, such as boiling point, viscosity, surface tension and heat of vaporization.
13. Write and balance a chemical equation if given the names or formulas of each reactant and product.
14. Define and calculate: theoretical yield, actual yield, percent yield and limiting reactant; and determine which reactant is the limiting reactant.
15. Classify compounds according to the type: salt, strong acid, weak acid, strong base, weak base, strong electrolytes, weak electrolytes, non-electrolyte.
16. Identify the three major types of reactions that can occur in aqueous solution.
17. Distinguish between exothermic and endothermic chemical reactions and physical processes.
18. Perform calculations related to the energy changes that accompany chemical reactions and physical processes.
19. Use the Ideal Gas model and equation to describe the behavior of a gas or a mixture of gases.

Final Comment on Learning Chemistry

The study of chemistry can be exciting and rewarding when there is a joint effort among students and instructors to continually improve learning. The General Chemistry program is designed to give you every opportunity to master fundamental chemical concepts and to demonstrate your mastery while earning a good grade.

Learning chemistry can be a challenge: you are confronted with a new language (terminology and symbolism) and you must synthesize new ideas while integrating your previous understanding of basic math and science. Success is a matter of exposure and practice, as any successful chemistry student will tell you.

Take advantage of all opportunities provided for your study of chemistry: the text, lecture, computer software, lab, help sessions, and office hours with your instructor. To earn a good grade, you must apply yourself in all of these areas. Your goal is to understand the material well enough to answer questions, both numerical calculations and questions that test your conceptual understanding, with an ease that comes from familiarity with a subject. Read your text and stay about one half of a chapter ahead of your lecturer.

After each lecture go back to your book to reinforce things that were unclear in class. This is important -- **do some chemistry every day**. Study of your text and attention in class will be most effective if you work with chemistry in small sessions, as opposed to "cramming" right before an exam. Do not expect that just because you go to class and listen that you are learning. You must explore chemistry on your own to make the subject a part of your working knowledge.

Learning is hard work. Make the effort to put in the time necessary to understand chemistry and you will be rewarded. If you fall behind seek help immediately. See your instructor as soon as you feel insecure with the material you are learning. It is better to address issues when they arise, before they become an obstacle to your progress.

Specific CH 1010 Section Information

Sec	CRN	Days	Time	Room	Instructor
1	10126	MWF	09:05-09:55	Newman 104	Ennis
2	11165	MWF	11:15-12:05	Newman 104	Cooper
3	15885	MWF	12:20-01:10	Daniel Expansion G66	Taylor
4	11168	TR	09:30-10:45	Newman 104	Choi
5	11174	TR	11:00-12:15	Newman 104	Choi
6	17884	TR	12:30-01:45	Kinard 201	Williams
400	11175	online	asynchronous		Schvaneveldt

Instructor Contact Information

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Dr. Sooyun Choi	sooyunc@clemson.edu	Hunter 269
Prof. Destinee Cooper	dkjohns@clemson.edu	Sirrine 262
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Dr. Stephen Schvaneveldt	schvan@clemson.edu	Hunter 275
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Dr. Trey Williams	cbw2@clemson.edu	Hunter 261